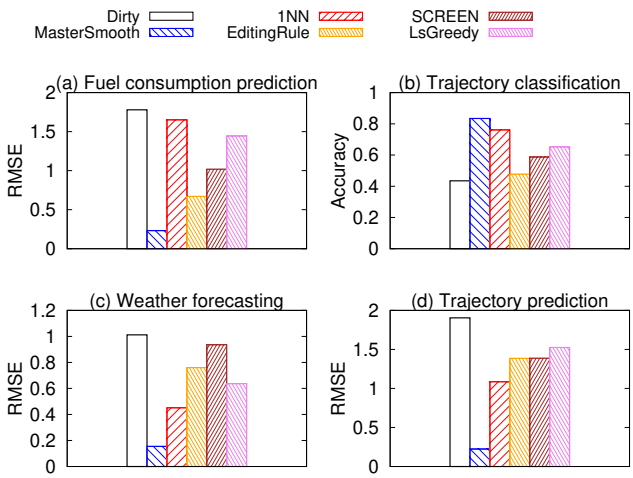

Figure 18: Applications with or without data cleaning

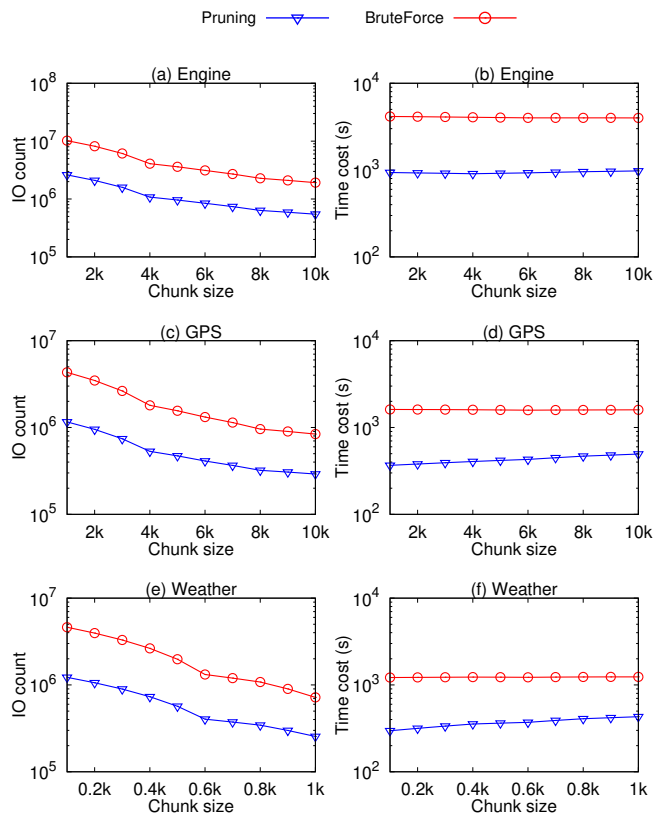

Figure 19: Performance of neighbor search in master data (IO count denotes the IO times of reading chunks from disk)

A APPLICATION EVALUATION

In this section, we share the real-world experiences of applying our proposal with master data in our industrial partners and public benchmark. Time series prediction is the most common application, including fuel consumption prediction [14] and wind turbine icing forecasts. It is also crucial for T-BOX in classifying dump truck trajectories. However, data quality issues make predictions and classifications on raw data unsatisfactory.

Figure 18 shows these applications without/with different smoothing methods. For prediction, we use LSTM [21]. For classification, we use DTW [38] as the distance measure and 1NN [34] as the classifier. (1) For fuel consumption prediction, Tianyuan Inc. aims to forecast vehicle fuel use for efficiency purposes [14] (Section 5.1.1). Compared to other cleaning methods, our method achieves the lowest prediction RMSE, as shown in Figure 18a. (2) For dump truck trajectory classification, Weichai Power requires accurate trajectory categorization (Section 5.1.2). Raw data errors make direct classification unreliable, but applying MasterSmooth significantly improves accuracy, reaching nearly 90% (Figure 18b). (3) For weather forecasting, Goldwind Co. predicts conditions for wind turbine icing early warnings (Section 5.1.3). Our method achieves an RMSE only half that of 1NN, thanks to superior cleaning performance (Figure 18c). (4) For taxi trajectory prediction over a public dataset [53] (Section 5.2.1), time series cleaning substantially enhances prediction performance, with MasterSmooth achieving the lowest RMSE among all methods (Figure 18d).

B PROOFS

B.1 Proof of Proposition 1

Proof. The sum of distances is easy to check in polynomial time, i.e., the problem is in NP. To prove the NP-Hardness, we show a reduction from the decision version of the Traveling Salesman Problem (TSP), one of Karp's 21 NP-Complete problems [25].

We first introduce the decision version of the TSP problem: Given an undirected weighted graph $G = (V, E)$ with n vertices and their pairwise distances, the problem is to decide whether there exists a path of total weight less than or equal to L, which visits each vertex exactly once. In addition, if all the vertices are in a plane (i.e., the distances satisfy the triangle inequality), existing studies also prove that the TSP problem is still NP-Complete under l_1 and l_2 distances [18, 22, 33].

Given an instance of the TSP problem in a plane, we then construct an instance of our Master Data Ordering Problem with 2 dimensions: For each vertice $v_i \in V$, we use $(v_i[1], v_i[2])$ to denote its coordinates, and then construct a tuple $y_i = (v_i[1], v_i[2]) \in Y$ corresponding to v_i. Therefore, the distances between vertices v_i, v_j are also equal to the distances between tuples y_i, y_j.

Following Problem 2, the decision version of our problem is to decide if there exists a master data order with the sum of the distances less than or equal to K. Let $K = L$, we prove that these two instances are equivalent. Note that the distances of two instances are equal, to find a path of total weight less than or equal to L in TSP problem, we then order the tuples following the order of the vertices in the path. Therefore, the path in TSP has total weight less than or equal to L, if and only if there exists an order of tuples with the sum of the distances less than or equal to K. To conclude, the NP-completeness of the decision problem is proved. □

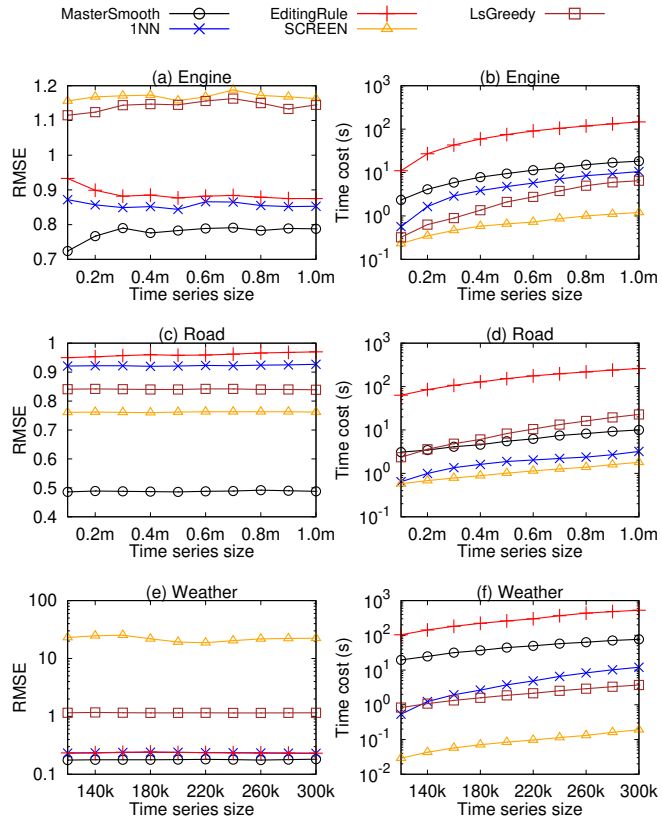

Figure 20: Performance of smoothing query by varying time series data size n

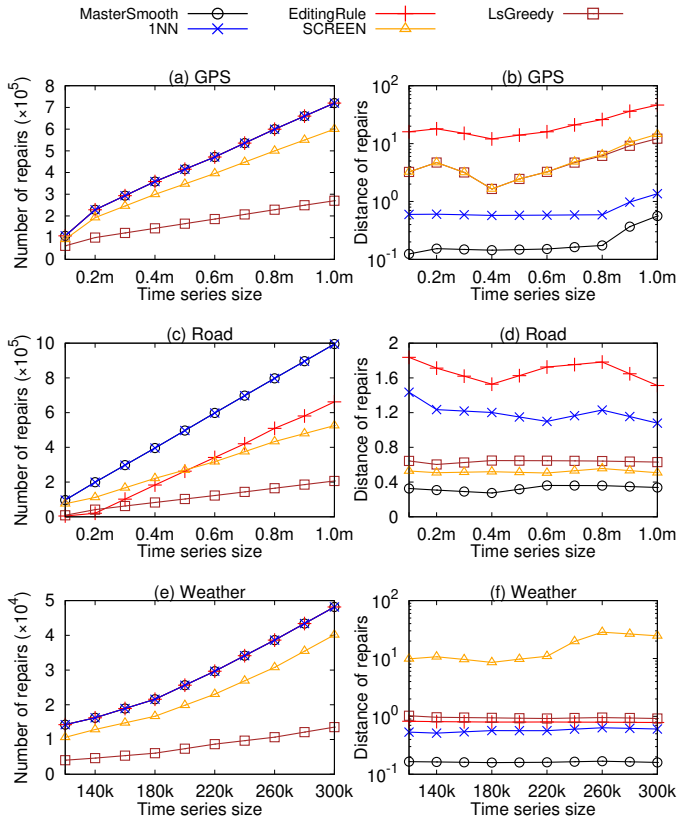

Figure 21: Repair cost of smoothing query by varying time series data size n

B.2 Proof of Lemma 2

Proof. First, we consider the distance between x and y in the j-th dimension. When $Y_i^{min}[j] < x[j] < Y_i^{max}[j]$, $x[j]$ is covered by the range of $Y_i[j]$. Thereby, $\theta_j(x, Y_i) = 0$ in this case. When $x[j] < Y_i^{min}[j]$, $x[j]$ is out of the value range of the j-th dimension. Then the value closest to $x[j]$ is $Y_i^{min}[j]$, which gives the minimum distance $|x[j] - Y_i^{min}[j]|$. Similarly, we could get the distance bound of the j-th dimension when $x[j] > Y_i^{max}[j]$. In summary, we have

$$|x[j] - y[j]| \geq \theta_j(x, Y_i), \quad \forall y \in Y_i, \quad 0 \leq j \leq l$$

The distance $\delta(x, y)$ is thus bounded by $\theta(x, Y_i)$, for $y \in Y_i$.

$$\delta(x, y) = \sum_{j=1}^{l} |x[j] - y[j]| \geq \sum_{j=1}^{l} \theta_j(x, Y_i) = \theta(x, Y_i)$$

□

B.3 Proof of Proposition 3

Proof. Suppose that there exists a neighbor y^* of x in cluster Y_i, i.e., $\delta(x, y^*) \leq \delta_\kappa$. On the other hand, according to the Lemma 2, $\forall y \in Y_i, \delta(x, y) \geq \theta(x, Y_i)$. It follows $\delta(x, y^*) \geq \theta(x, Y_i)$. We thus obtain $\theta(x, Y_i) \leq \delta_\kappa$. □

B.4 Proof of Proposition 4

Proof. The proposition can be proved similarly following the proof of Proposition 3. □

B.5 Proof of Proposition 5

Proof. First, assume $\Delta(X, X') < \Delta(X_{1:u}, X''_{1:u}) + \Delta(X_{u+1:n}, X''_{u+1:n})$. Since $t_{u+1} - t_u > \omega$, it implies $\Delta(X_{1:u}, X'_{1:u}) + \Delta(X_{u+1:n}, X'_{u+1:n}) < \Delta(X_{1:u}, X''_{1:u}) + \Delta(X_{u+1:n}, X''_{u+1:n})$. That is, at least one of $X''_{1:u}$ or $X''_{u+1:n}$ is not the minimum modification, which leads to a contradiction. Then, assume $\Delta(X, X') > \Delta(X_{1:u}, X''_{1:u}) + \Delta(X_{u+1:n}, X''_{u+1:n})$. Given such a X'', it implies that X' is not a minimum modification of X, again a contradiction. Hence, the equality must hold. □

B.6 Proof of Proposition 6

Proof. Note that the smoothing tuples in our algorithm are selected from the valid modification set V_i. Thereby, the modification x'_i of x_i exists if and only if V_i is not empty. We discuss two cases of window W_i.

If $W_i = \emptyset$, the smoothness constraint $\mathcal{S}$ of x_i is naturally satisfied. Therefore, we can always find a candidate from $\text{Nei}(x_i)$ in the master data to modify x_i safely.

If $W_i \neq \emptyset$, then the candidate modification set C_i involves $\text{Nei}(x'_j)$, for $x'_j \in W_i$, i.e., $x'_j \in C_i$ as well. Due to the streaming

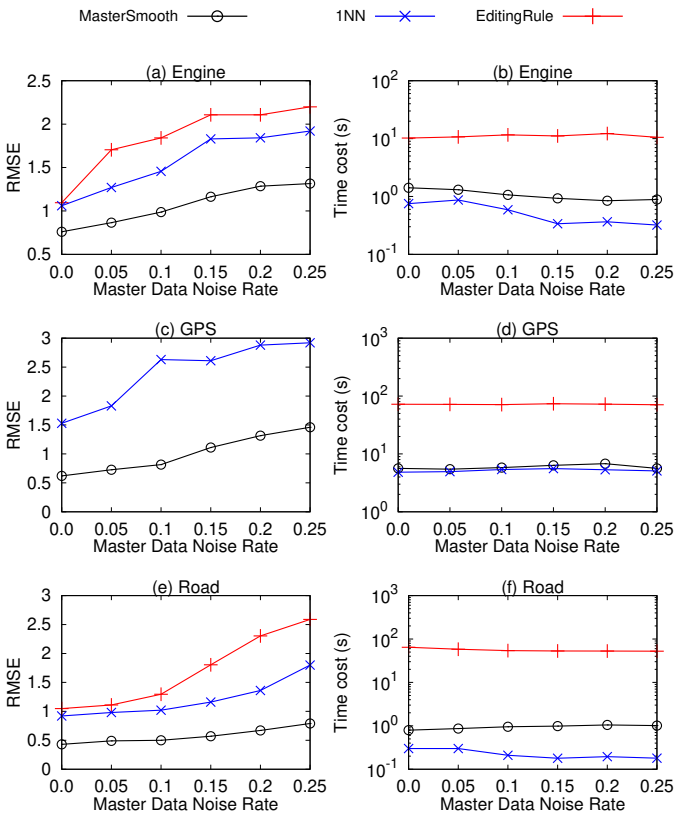

Figure 23: Performance of smoothing query by varying noise rate of master data (EditingRule shows significantly higher RMSE in (c) and is omitted)

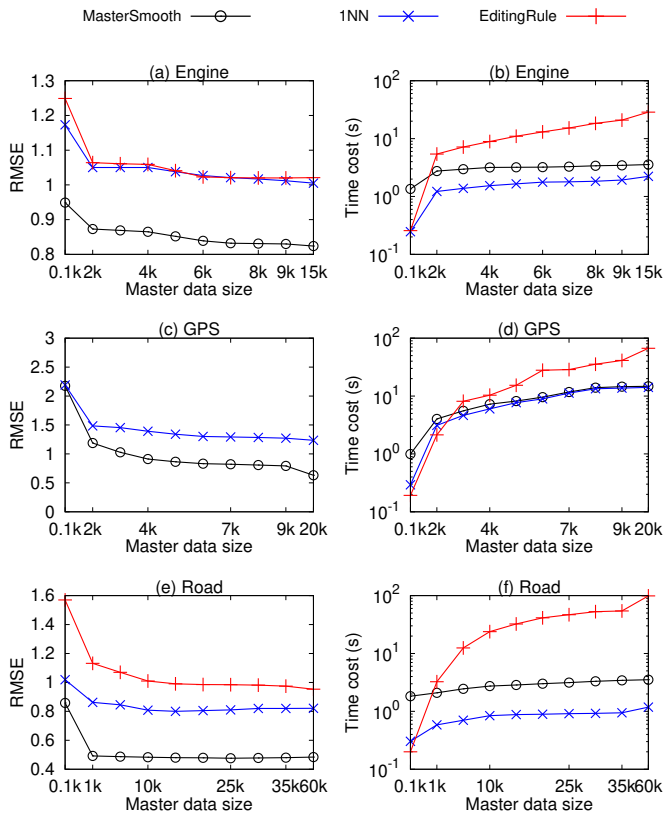

Figure 22: Performance of smoothing query by varying master data size m (EditingRule shows significantly higher RMSE in (c) and is omitted)

computation, x_j is smoothed before x_i. That is, x'_j in the window satisfies the smoothness constraint with each other, referring to the previous smoothing steps. Therefore, there always exists $x'_j \in W_i$ as the valid modification of x_i. □

B.7 Proof of Proposition 7

Proof. For window size $\omega = 0$, the temporal smoothness of the time series is no longer considered. The window W_i of each tuple x_i is always empty and the candidate set C_i is $\mathrm{Nei}(x_i)$. Since the modifications no longer affect each other, we have $V_i = C_i$. The one with the minimum distance to x_i in V_i returned by Algorithm 2 is indeed the optimal solution. □